

Report Milestone 2.1: Audio Processing

1. Obtaining Data and Query Preparation

We used the database raw_30s_audio-21 (low quality version) which contains 554 songs. From there we randomly selected 20 songs and created the versions original, noise, coding and mobile.

2. Database Preparation

We chose the combinations of the values [5, 11, 21] for κ and [3, 7, 15] for τ . (Given below as tuples of (κ, τ) .)

We additionally tested three extra combinations [(5, 9), (5, 11), (7, 7)] to explore different settings. As you can see below, they did not yield better results though, as the original settings already produce very good results.

3. Retrieval Experimentation

The reference implementation uses TP only; however, precision, recall, or F measure could be chosen, too. Using only TP is not ideal because it's biased toward dense regions. Therefore, it's better to use a scoring method which takes FP and FN into consideration. We used F measure because we experienced the best results with it and it uses precision and recall.

Results: (hits/queries)

	original	coding	noise	mobile	all	time/query
(5, 3)	100%	100%	95%	95%	98%	7,419s
(5, 7)	100%	100%	100%	95%	99%	7,05s
(5, 15)	100%	100%	100%	95%	99%	6,94s
(11, 3)	100%	100%	95%	95%	98%	6,871s
(11, 7)	100%	100%	100%	95%	99%	6,886s
(11, 15)	100%	100%	100%	95%	99%	7,041s
(21, 3)	100%	100%	95%	95%	98%	7,004s
(21, 7)	100%	100%	95%	95%	98%	7,0s
(21, 15)	100%	100%	100%	95%	99%	7,0s

(5, 9)	100%	100%	95%	90%	96%	6,894s
(5, 11)	100%	100%	95%	90%	96%	6,799s
(7, 7)	100%	100%	95%	95%	98%	7,9s

Each of our queries takes on average around 7s to execute and can have an accuracy of 99% with the right settings.

The best results are with the setting (11, 7), as it takes the least amount of time. But the differences are so small that (5, 7), (5, 15), (11, 15), and (21, 15) are also good choices.

Track 117221 is the hardest to identify because its first 30 seconds contain very few prominent features. This is especially difficult for the noise and mobile queries, where the added noise dominates the signal. This explains why these versions have worse results.

We implemented the following optimizations in order to speed up the retrieval tests:

- Adjusting the Delta parameter in `compute_matching_function` to shorten the execution time.
- Increasing H in `compute_spectrogram`, which reduced computation by lowering time resolution (fewer frames). Doubling it has almost no change in the results but increasing it by a higher factor would lead to fewer hits.
- Replacing `match_binary_matrices_tol` with a version utilizing numba jit-compilation (see `match_binary_matrices_numba`). This has a big impact on execution time.